

# Walnut Works

# Welcome to Walnut Works

Kabir Hussain MA and Vicky Hussain MA PGCE are both artists with individual art practices.

Walnut Works was established in 2019. The idea had been formulated many years earlier to utilise Kabir's 30 years of experience working in commercial fine art foundries with prominent 20th and 21st century artists and Vicky's experience in schools and as a founder member, gallery organiser, and curator of five bronze shows at Greyfriars Art Space in King's Lynn.

Both artists have used the medium of bronze in their own practice as sculptors since 1985

Kabir wanted to encourage artists and give them the opportunity to work directly with the processes of fire 'cire perdue' or lost wax casting mainly utilising direct burning out technique used in his own sculpture practice.

Vicky has organised group wax modelling workshops for all ages and practices nationwide using her wax to bronze kits.

Over the years we have developed a joint teaching facility offering a one-to-one

course of consultation and group workshops for adults and children nationwide.

## The guest artist program

The guest artist program initially arose when we approached artists with long-standing practices and reflected on how they may benefit from working directly in bronze. We offered this unique opportunity to realise a body of work which due to the technical and financial limitations would ordinarily be difficult for them to achieve. Unlike a formal foundry environment, a Walnut work residency serves as an extension of studio practice, with guidance allowing them to investigate materials and processes in a hands-on manner.

The artist arrives at the studio with a pre-formulated idea and processes these ideas through an explorative phase of handling the materials to familiarise themselves with the casting procedure; through interaction with grog, luto, and plaster.

This technique was practised at the Royal College of art having been introduced by the Angelone Brothers from Rome.

Cire Perdue extends the artist's vocabulary

and establishes a bridge with the Renaissance Masters. After the physically grounding experience of making the investment mould - there is a significant moment of the big reveal - breaking open the cast after the metal has been poured and cooled.

The metalworking stage allows for a greater understanding of surface and form of the piece. Patination completes the cycle giving the artist a bond with their own work and a sense of satisfaction with their own endeavours.



# Bronze sculpture from the beginning

Modern historians believe that setting up stones around a fire enabled early people to extract metals from rocks. The earliest bronze artefacts were found in the Middle East and East Asia approximately 7000 years ago. Bronze artefacts were also found at this time in parts of Serbia. The historical Bronze Age in Britain started approximately 5000 years ago. (FIG. 1)

Bronze was discovered by mixing two weak metals (which were absolutely no good for making tools because they had no tensile strength.) Copper and tin makes a stronger alloy material; bronze. Copper has been mixed to make other bronze alloys; lead and arsenic as well as tin. Lead until very recently was used in a bronze alloy. (FIG. 2)

Tools and weapons, armour and fittings for chariots as well as musical instruments were made of bronze in the ancient world.

In the later Bronze Age cannons were invented in East Asia. Sparks in bronze cannons do not cause explosions or erode in water. Bronze is still

used today for electrical components, weapons, anchors and nautical equipment.

In in her book 'Hoards Hidden History' published by The British Museum Press Eleanor Ghey writes : "Trade spread the knowledge throughout the ancient world. Many later Bronze Age hordes were composed of scraps of objects to be recycled as raw material for metal work and have traditionally been referred to as founders hordes. These groups tend to include ingots and unfinished items. It is likely that we have here the evidence for Smiths at work this is not to say these deposits were not seen as having a special character. In some societies the Smith is seen as the possessor of supernatural powers".

As casting methods improved small figurative sculptures appear. Bronze expands when heated and fills all cavities in the mould which gives the surface of the material texture.

The earliest bronze figurative sculptures appear in Mohenjo-daro now modern Pakistan circa 2500 BCE.

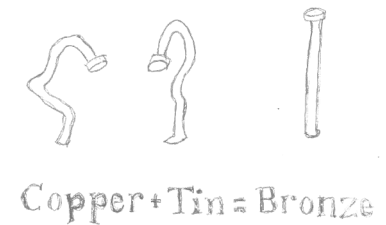


FIG. 2 - COPPER + TIN = BRONZE



FIG. 1 - FIRE

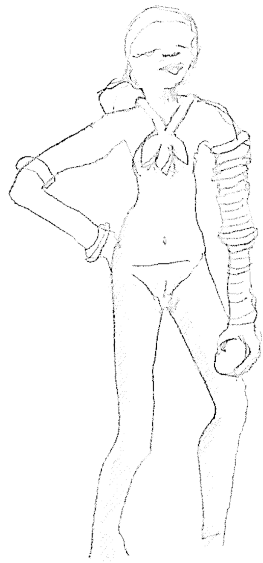


FIG. 3 - FEMALE SCULPTURE  
MOHENJO-DARO

Traces of this early civilisation were found throughout India and modern day Pakistan. A young female and the head of a man were discovered by archaeologists in the 1920's. They were kept together for many years but separated during the partition of British India - the young female moving to a Museum in Pakistan and the male head staying in Delhi. (FIG. 3)

The ancient Greeks developed the casting process to make full sized figurative sculptures. Few Greek full-sized statues survive, the bronze being repurposed and recycled. Romans emulated the Greeks and bronze was used for all sizes of statues and statuettes reproducing sculptures using moulds to make multiple figures and figurines. (FIG. 4)

In the European mediaeval period bronze statues and decorations would have been found mainly in a religious contexts, churches and palaces.

During the Renaissance artists started to rediscover Roman and Greek sculpture and architecture and create their own large-scale sculpture. The sculpture was patronised by powerful families in Europe and the Italian kingdoms. The horses of Saint Marks' arrived in Venice in 1204 after being looted from Constantinople during the fourth crusade. The crusaders who were equipped by the great Venetian shipbuilders took a

detour from the Holy Land (modern Palestine) and raided Constantinople (modern Istanbul) they used this loot to pay their debt to the Venice boat-builders. The payment included the ancient horses; originally part of an ancient stadium and were placed in the grand St. Mark's Square. (FIG. 5)

Rich and powerful families of the Italian kingdoms became the patrons of artists fuelling the rediscovery of bronze casting techniques throughout Europe. (FIG. 6)

In the 18th-century interest in Italian and Greek art and architecture was prominent in British culture. Wealthy young men, as a way of finishing their education, went on 'The Grand Tour' around the cultural cities of Europe. They bought souvenirs of art including bronze statues and statuettes back to fill the collections of stately homes and museums.

Patronage of large-scale public sculpture was a way of showing power in public spaces. The wealth of the British was often gained as a result of slavery, colonialism and the new technology impacting the Industrial Revolution. Public statues were made of important and powerful people. These were mainly of men who were generals, industrialist, politicians or royalty. Queen Victoria being an exception. Sculptures of her were placed throughout the British Empire. Women



FIG.5 - CRUSADER SHIP



FIG. 4 - THE BUST OF THE EMPEROR  
HADRIAN, FOUND IN 1853 IN  
THE THAMES



FIG. 6 - DUKE COSIMO I COMMISSIONING  
CELLINI'S 'PERSEUS AND  
MEDUSA'



FIG.7 - QUEEN VICTORIA

en were often represented in sculpture as mythical creatures nymphs and fairies. Queen Victoria herself often bought her husband sculptures of semi naked woodland nymphs bordering on erotica. Sculptures of nymphs semi-clad nubile figures of myths and legends were very popular with the Victorian public and were notoriously prudish covering furniture legs so as not to be indecent. (FIG. 7)

The statue commemorating the slave owner Edward Coulson, a 17th century slaveowner was toppled in 2020 by protesters into the Bristol Bristol harbour. There is a debate today on how we should view and show these works. (FIG. 8)

At the beginning of the 20th century foundries were set up to produce war memorials to honour the death of the First World War dead. There were huge shifts in the arts with modernism in Europe.

Postwar Britain – British art and design exploded in the 50s and 60s with art colleges reflecting the new modernist teaching. Britain was the centre of the Art World.

The festival of Britain introduced new ideas of sculpture architecture and textiles ceramics and music. Colleges were set up reflecting this new patronage from the postwar government encouraging the rebuilding of

industry and infrastructure. The Arts Council was established in 1946 to support and develop arts and culture in Britain.

Architecture and town planning often included public modern sculpture. Harlow sculpture Town architect Sir Frederick Gibberd designed areas where people could step out of their front door and find a Barbara Hepworth sculpture. Originally built for people not cars, sculpture and landscape informed his designs and planning. Students from Harlow College could take a few steps away from the college( now and be surrounded by bronze sculpture in the water gardens. Liz Frink, Henry Moore, Rodin. In the area walking towards the marketplace there is a a Ralph Brown, a Lynn Chadwick, amongst others. Sadly the college has been knocked down with the sir Frederick Gifford town hall.

‘Old Flo’ sold by Henry Moore at a modest price to Tower Hamlets Council to say thank you to the people of the East End after the war and placed outside council flats.(since demolished). Beloved by local children and tenants it was moved to Yorkshire Sculpture Park after the council tried to sell it causing a scandal. Recently it was moved to a site in Canary Wharf surrounded by the financial district. Worth millions and apparently expensive to maintain, it

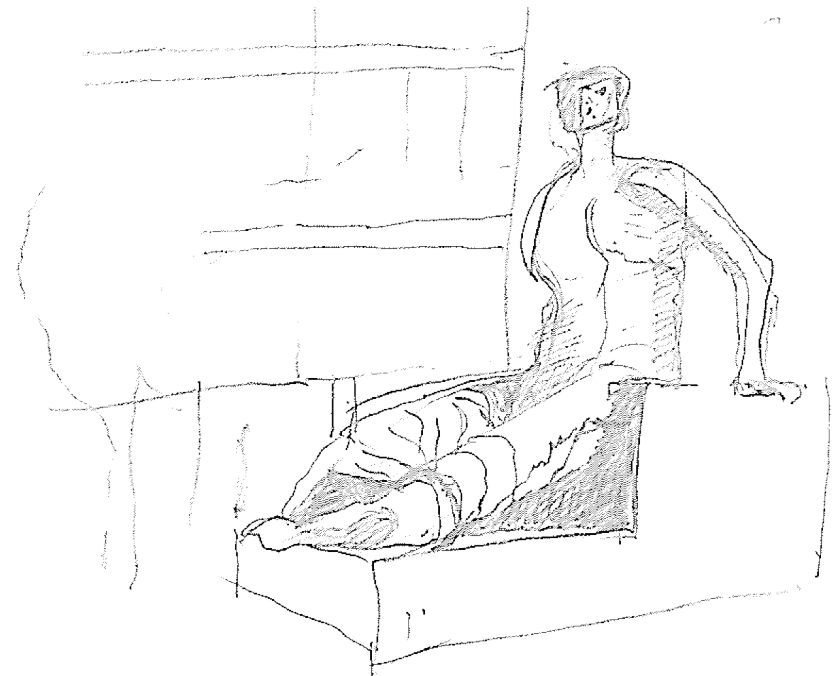


FIG.8 - TOPPLING COULSON

is deemed a temporary safe site.

Artist tutors from the 40's, 50's, and 60's have influenced the last decades of the 20th century and have in turn influenced modern bronze sculptures such as Gavin Turk, Anish Kapoor, Grayson Perry, Antony Gormley, Anya Gallachio, Louise Bourgeois to name a few. <sup>(FIG. 9)</sup>

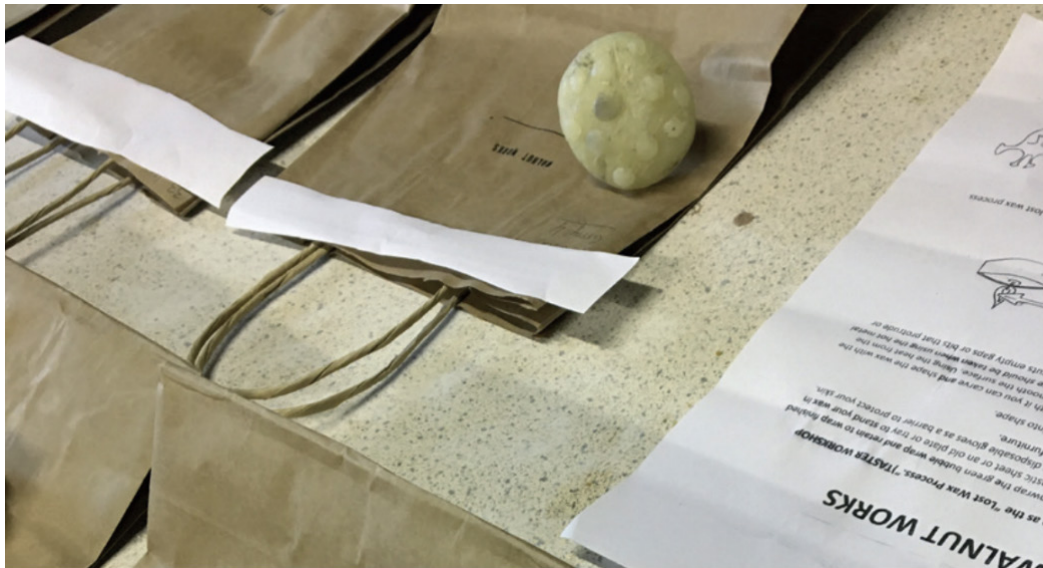
Today college foundries are disappearing. Commercial arts foundries often use ceramic shell to process sculpture which is cost effective in mechanised foundries. The lost wax process using Luto is at risk of becoming obsolete in college foundries this hands on method used by Renaissance founders is in danger of becoming a heritage craft. Working with artists directly with this process we wish to share our knowledge and skill with with our resident artists to experience for themselves.





# The bronze casting process

*Kit bags arrive at the studio, and are labelled with the artist's details.*



*We roll out runners (thick ones) and risers (thin ones). The thick ones will leave a wide gap in the mould after it's been fired in the kiln for the metal to flow. The thin ones leave a space for the air to escape.*





*Kabir brushes on the first coat of the mould; a mixture of herculite plaster and fine casting plaster. This coat has to be very strong to pick up the details on the wax.*



*A mix of plaster and water to make the mould.*





*To make a plaster for the second coat of the mould, Vicky is crushing waste plaster from the fired broken moulds.*

*The wax sprue system is being encased in a second coat of luto and fine casting plaster; called grog.*

*Kabir builds up the second layer of 2 parts luto to 1 part fine casting plaster.*

*We take the finished moulds to the foundry kiln.*



*Kabir puts the moulds into the kiln to fire for five days. The wax burns out to leave a space to pour the hot metal.*



*The moulds are cooked and still warm from the kiln. The wax has melted away and the mould is fragile. Kabir wraps some scrim dipped in plaster around the moulds to strengthen them for the hot metal.*



*The melted metal ready to be poured into the moulds.*



*Meanwhile the bronze ingots are being heated up in the furnace. The bronze is heated up to 1080 degrees Celsius.*

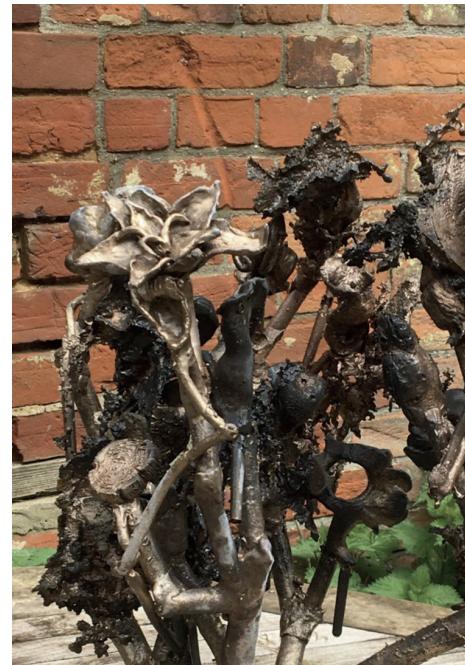




*The pour, and the top of the mould. You can see the wax has melted and burned away in the kiln - before and after the metal has been poured.*



*The moulds have cooled down and an axe is used to break the mould and find the bronze.*



*When the plaster is washed off the bronze. The runners and risers of sprue system are left with the small bronze sculpture attached.*



*Fettling the bronze.*





*Patination. Using chemicals to burn a colour the metal.*



*The finished small bronze sculptures.*



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